The VSPA firmware

NXP's DFE reference application, stripped to the one configuration this repository targets: **LA9310, single VSPA2 core, 16 AU, 1T2R, 25 MHz bandwidth, 30 kHz SCS** (61 subcarriers TX / 61 RX).

```
make -C firmware      # -> firmware/eld/dfe-la9310-1t2r-25m30k.eld
```

Builds natively with `fsvspacc`. No Docker, no Wine, no CodeWarrior.

> **This half is NXP's.** 61 files here carry `NXP Confidential` headers. Edit as little as possible; the simulation half is where our work belongs.

What is in here

`src/`	the application — TX and RX chains, antenna buffer management, host messaging
`include/`	its headers
`lib/`	`lib_DFE_Ref_vspa2.elb`, NXP's DSP library (the real iFFT, FFT, filters, NCO, QEC), plus its headers
`lcf/`	`LA9310_sp.lcf` — the memory map
`prebuilt/`	two object files that cannot be rebuilt on Linux (below)

The build is configured entirely by preprocessor defines:

```
-D__LA9310__ -DLA9310 -DVSPA2 -DSELF_CORE_ID=0
-DPROJ_CONFIG_OPTION=PROJ_1T1R_1CORE_LS_9310
```

The original DFE reference is a multi-core VSPA3/LA1224 project; the LA9310 build config is the single-core one, and it is the only one wired up here.

The two prebuilt objects

`prebuilt/` exists because two translation units cannot be compiled by the Linux toolchain. They are checked in as `.eln` objects, built by CodeWarrior on Windows:

object	why
`inter_core_msg.eln`	internal compiler error in backend codegen (`fscgg_vspa.cpp:282`)
`kernel_FFT/ifft_cp.eln`	`fsvspacc` cannot resolve IPPU function addresses at compile time

Their sources are still here (`src/inter_core_msg.c`, `src/kernel_FFT/ifft_cp.c`) — they just aren't in the compile list. If a later toolchain fixes either bug, drop the object from `PREBUILT_OBJS` and add the source to `LOCAL_C_SRCS`.

`src/startup_vspa2_modified.s` is a locally modified crt0; the stock one is in the SDK.

Linker warnings are expected

The link emits `W2003: found undefined input symbol` for a handful of symbols (`_gen_decx2_NTaps_align_asm`, `_bitRev_fft_4096_b0_b1_wrap_cp_insert`, ...). These are entry points for configurations this build does not use — 4096-point FFT, other antenna counts. They are never called on the 1T2R 25M/30K path, and the image is correct: it is **bit-exact against NXP's own reference build** across 18 vector/config cases, and matches MD5s captured from real LA9310 silicon. See `../docs/TESTING.md`.

Is it right?

Don't take the build's word for it — the whole point of `../cosim/` is to answer that question. `make verify-full` from the repo root runs the tiers.